# Initial assignment across the partner episode — all seeds

Seed 1 | Seed 2 | Seed 3 | Seed 4 | Seed 5 | Mean of 5 seeds

v1_matched | v2 | v3

MLP Ego-red assignment (%)

RNN Ego-red assignment (%)

RNN single-partner Ego-red assignment (%)

RNN no influence Ego-red assignment (%)

Round within partner episode

Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Red means ego takes red, partner takes blue. Blue assignment percentage = 100 − red percentage.

# Final assignment across the partner episode — all seeds

Seed 1 · Seed 2 · Seed 3 · Seed 4 · Seed 5 · Mean of 5 seeds

v1_matched | v2 | v3

MLP Ego-red assignment (%)

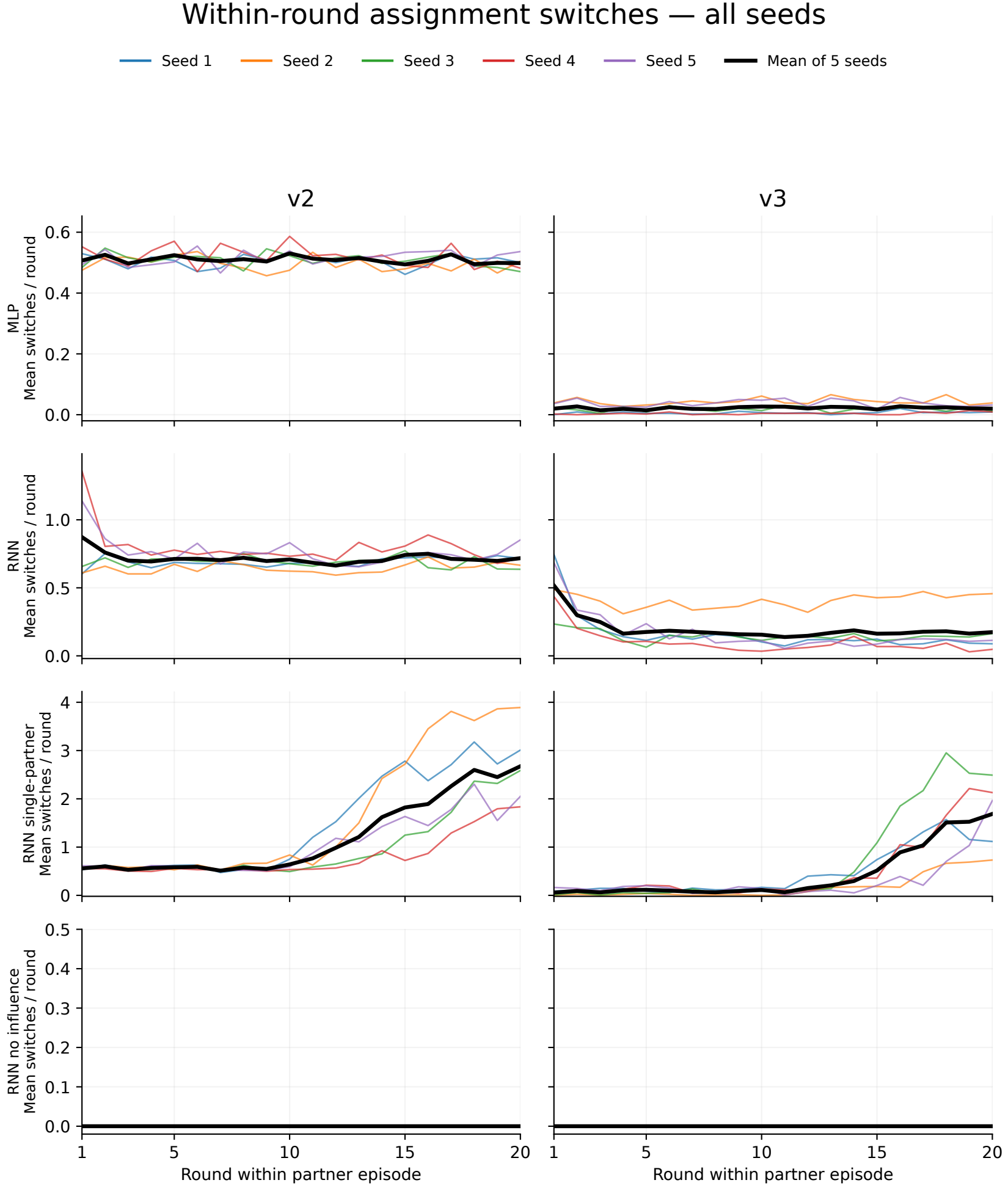
RNN Ego-red assignment (%)

RNN single-partner Ego-red assignment (%)

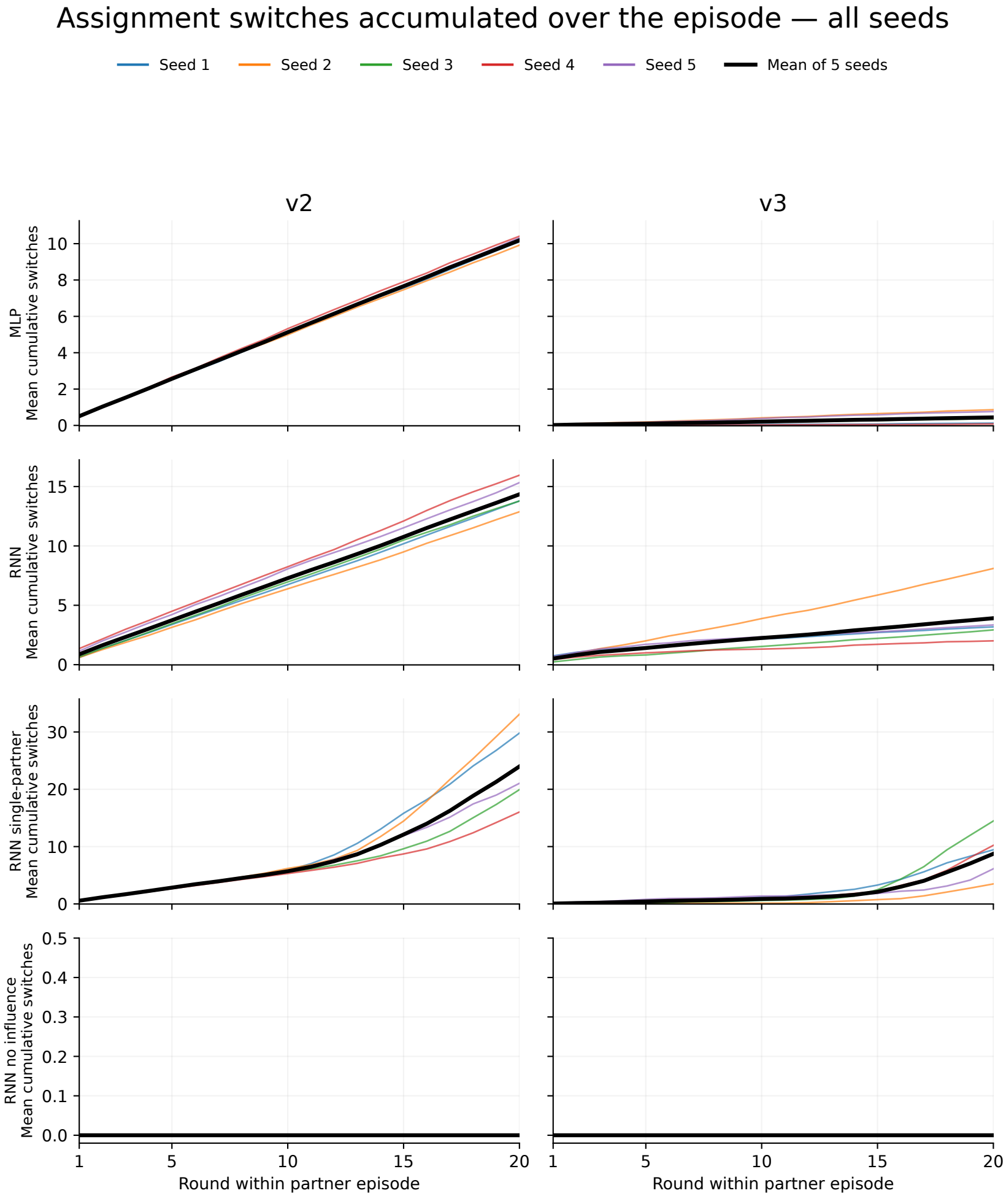
RNN no influence Ego-red assignment (%)

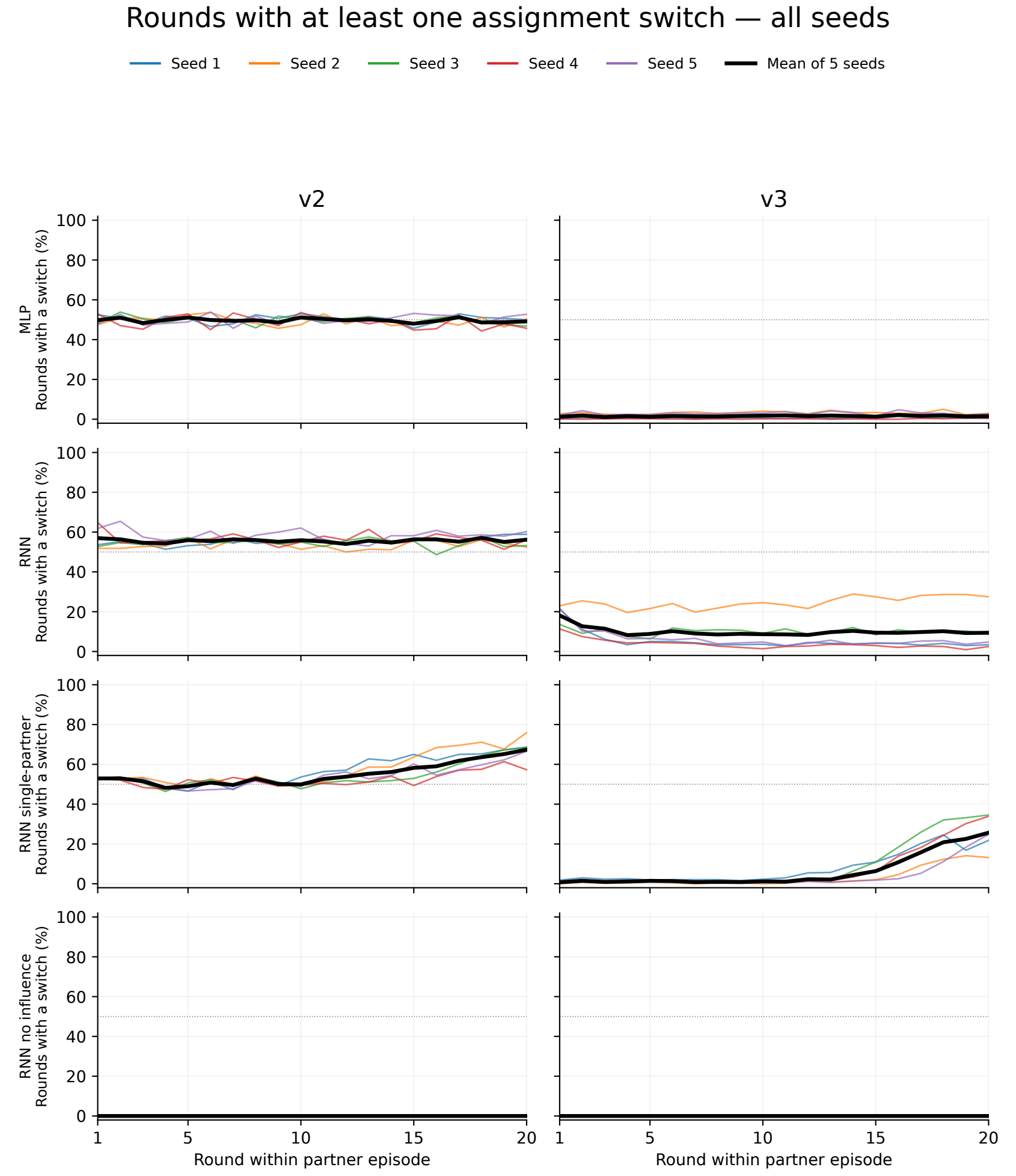
Round within partner episode

Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Red means ego takes red, partner takes blue. Blue assignment percentage = 100 − red percentage.




Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.


Assignment switches accumulated over the episode — all seeds
Seed 1
Seed 2
Seed 3
Seed 4
Seed 5
Mean of 5 seeds
v2
v3
MLP
Mean cumulative switches
RNN
Mean cumulative switches
RNN single-partner
Mean cumulative switches
RNN no influence
Mean cumulative switches
Round within partner episode
Round within partner episode
Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.




Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.

# Final assignment matches partner's faster task — all seeds

Seed 1 · Seed 2 · Seed 3 · Seed 4 · Seed 5 · Mean of 5 seeds

v1_matched | v2 | v3

MLP — Partner on faster task (%)

RNN — Partner on faster task (%)

RNN single-partner — Partner on faster task (%)

RNN no influence — Partner on faster task (%)

Round within partner episode

Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.

# Final assignments by partner ability

Partner faster at red (speed-matching ego-red target: 0%)
Partner faster at blue (speed-matching ego-red target: 100%)

v1_matched | v2 | v3

MLP — Final ego-red assignment (%)
RNN — Final ego-red assignment (%)
RNN single-partner — Final ego-red assignment (%)
RNN no influence — Final ego-red assignment (%)

0 20 40 60 80 100

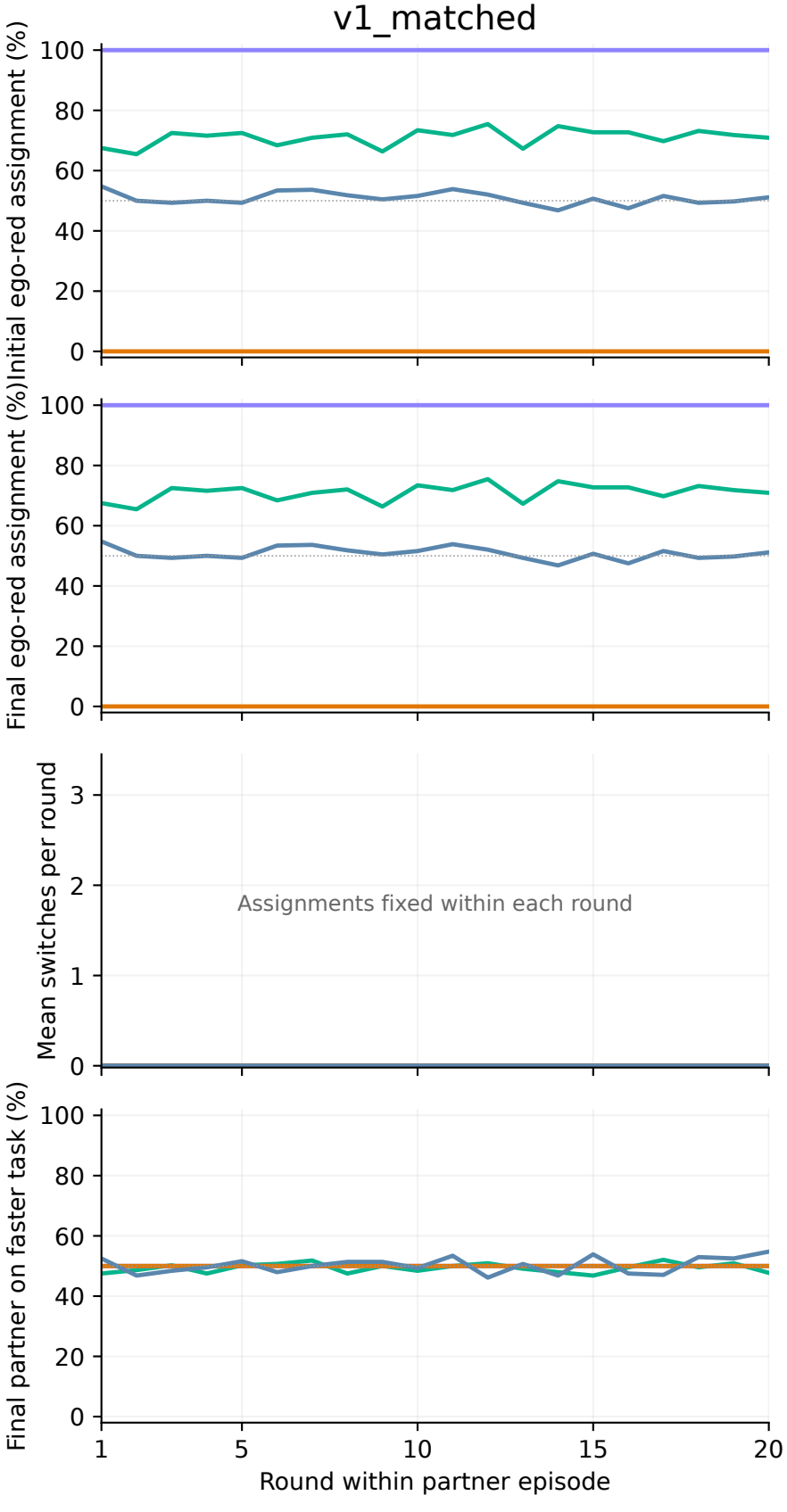
1 5 10 15 20
Round within partner episode

Thin lines: all five individual training seeds; thick lines: equal-seed means.
Partner color is complementary to ego assignment. Speed matching is a behavioral diagnostic, not a geometry-aware optimal policy.

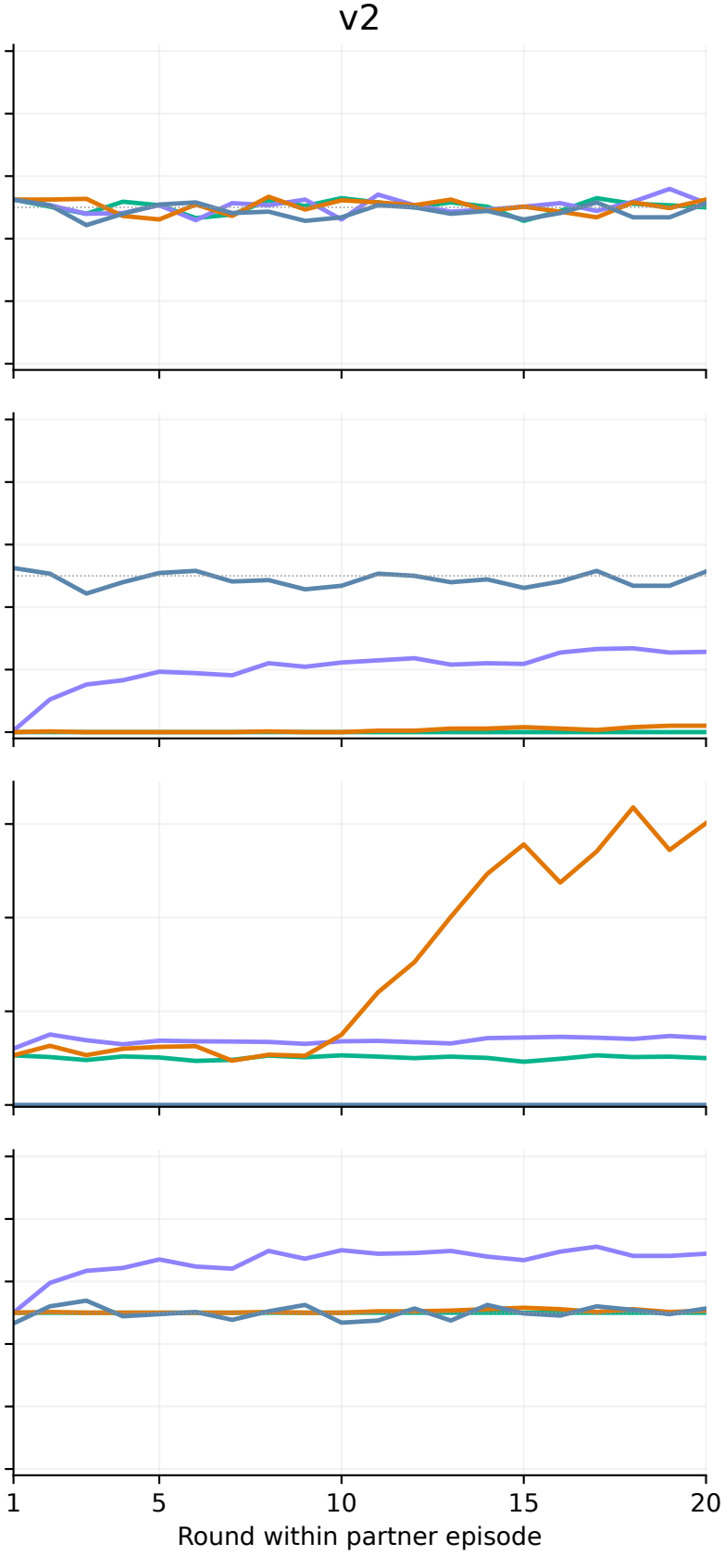
# Assignment behavior: training seed 1

MLP · RNN · RNN single-partner · RNN no influence

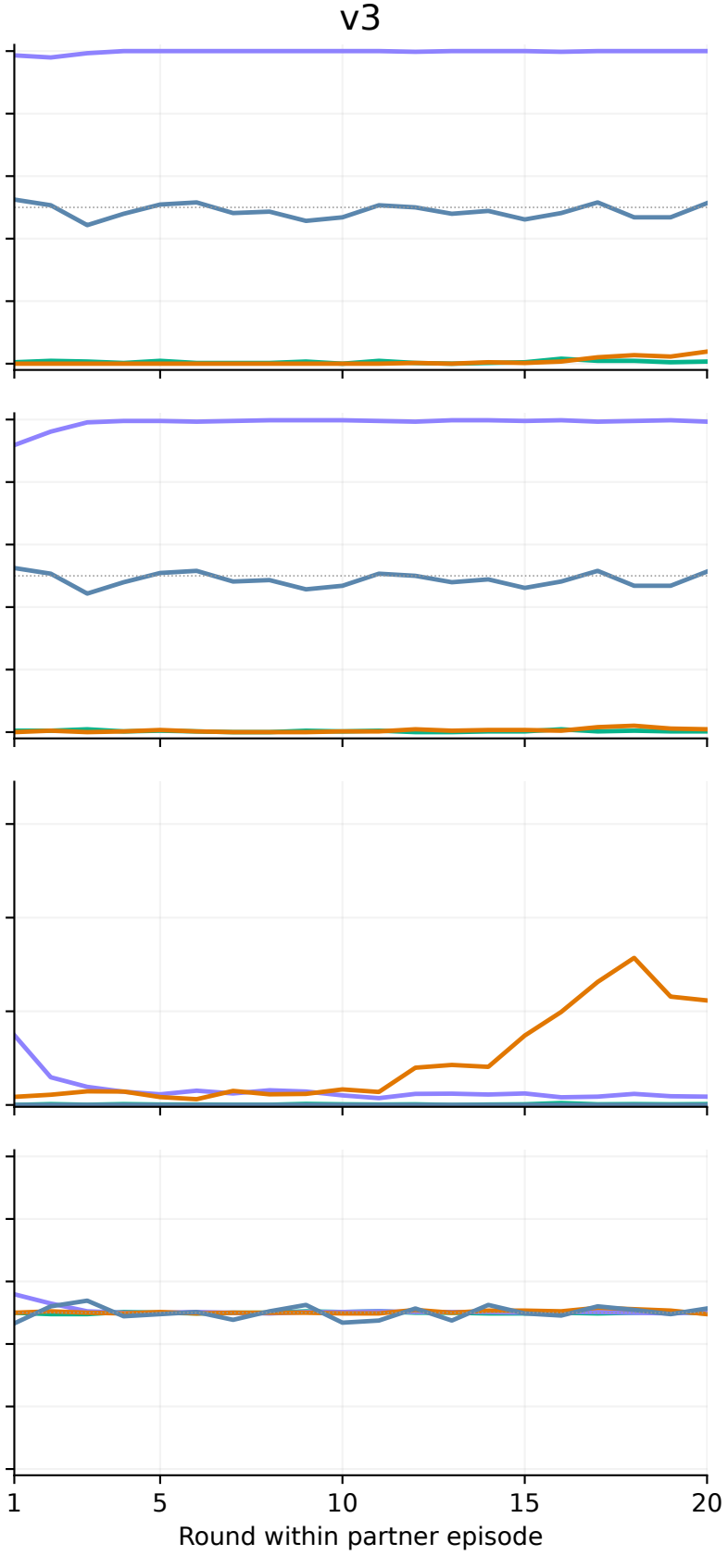
## v1_matched




## v2




## v3




Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

# Assignment behavior: training seed 2

MLP · RNN · RNN single-partner · RNN no influence

## v1_matched

## v2

## v3

Initial ego-red assignment (%)

Final ego-red assignment (%)

Mean switches per round

Assignments fixed within each round

Final partner on faster task (%)

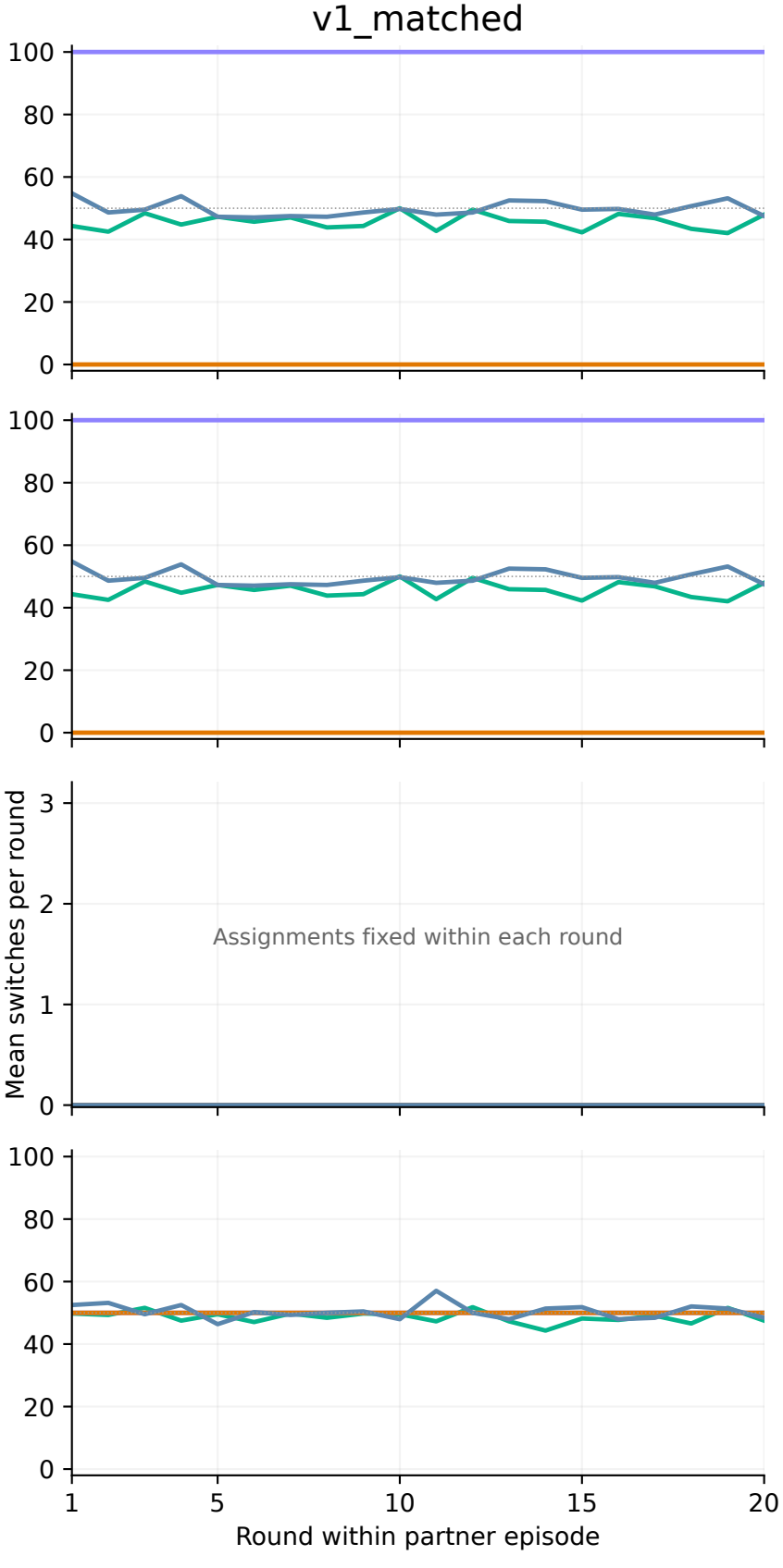
Round within partner episode

Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

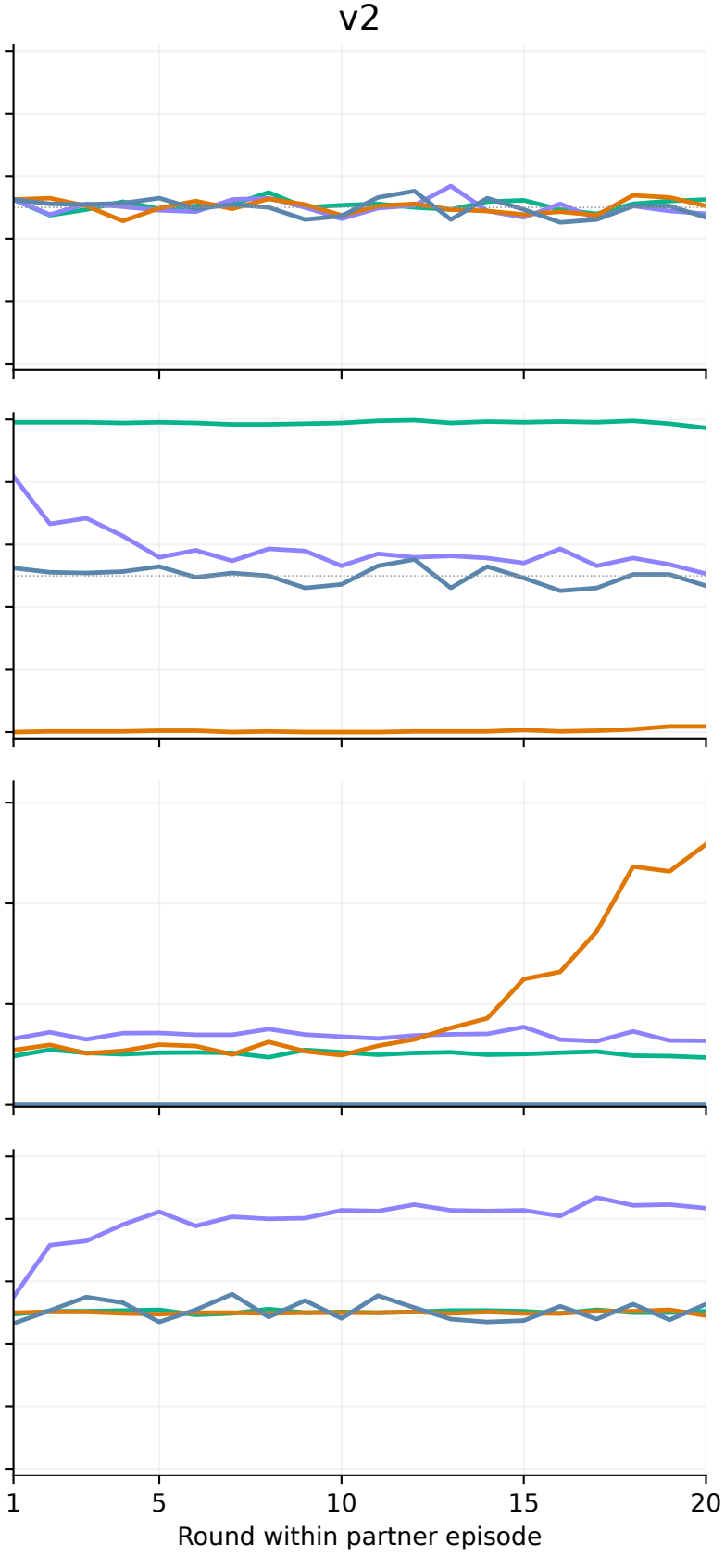
# Assignment behavior: training seed 3

MLP · RNN · RNN single-partner · RNN no influence

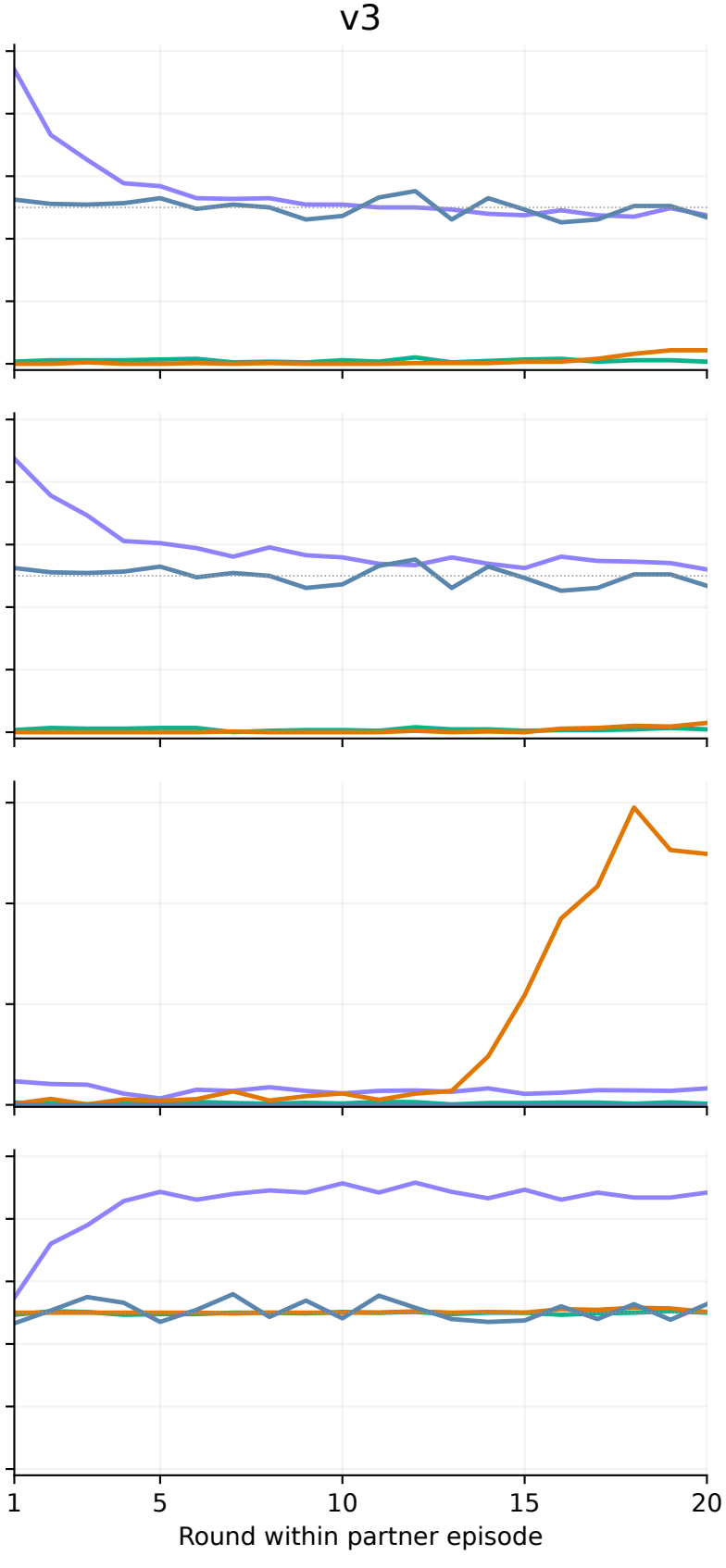
## v1_matched




## v2




## v3

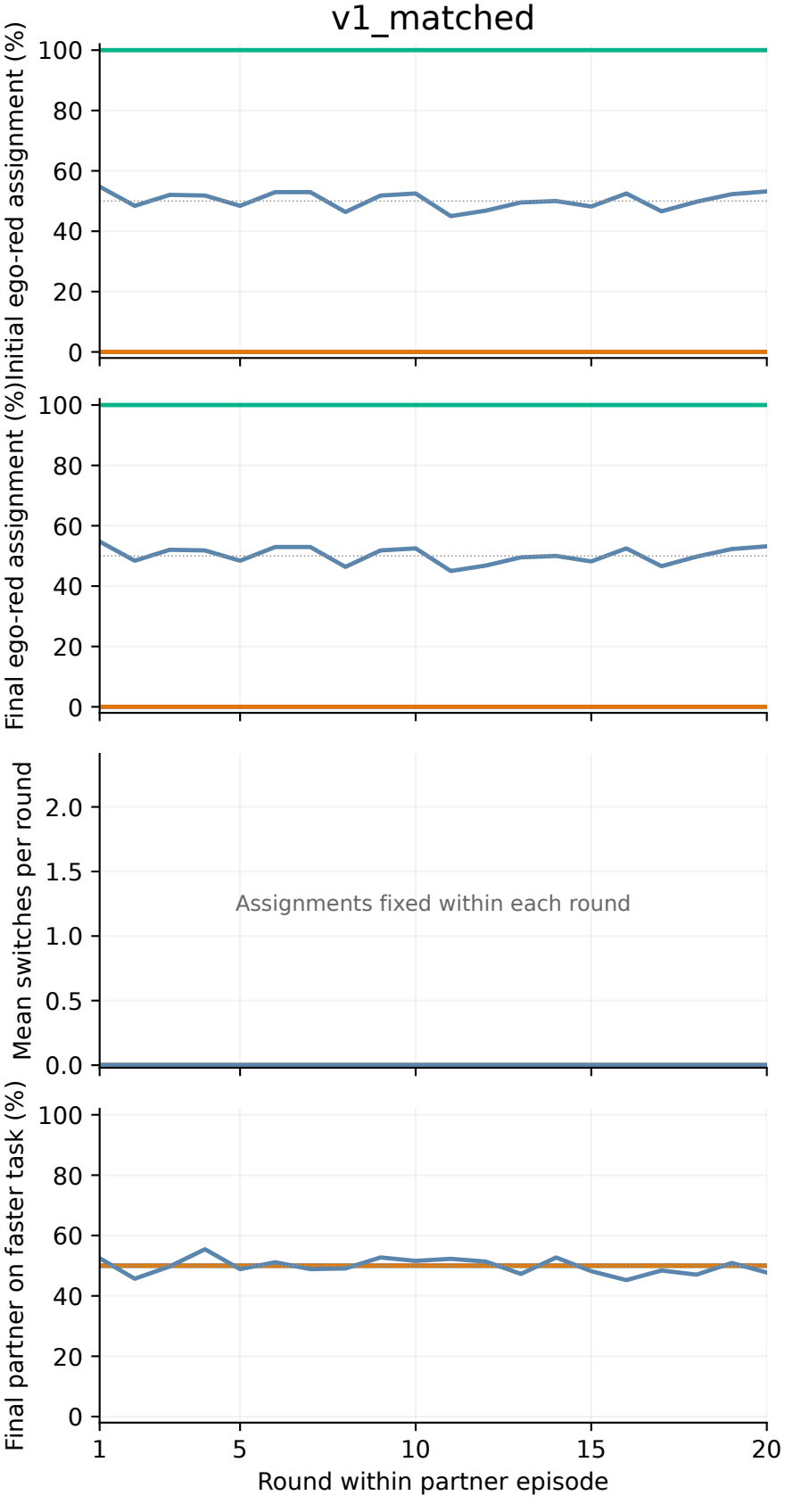



Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

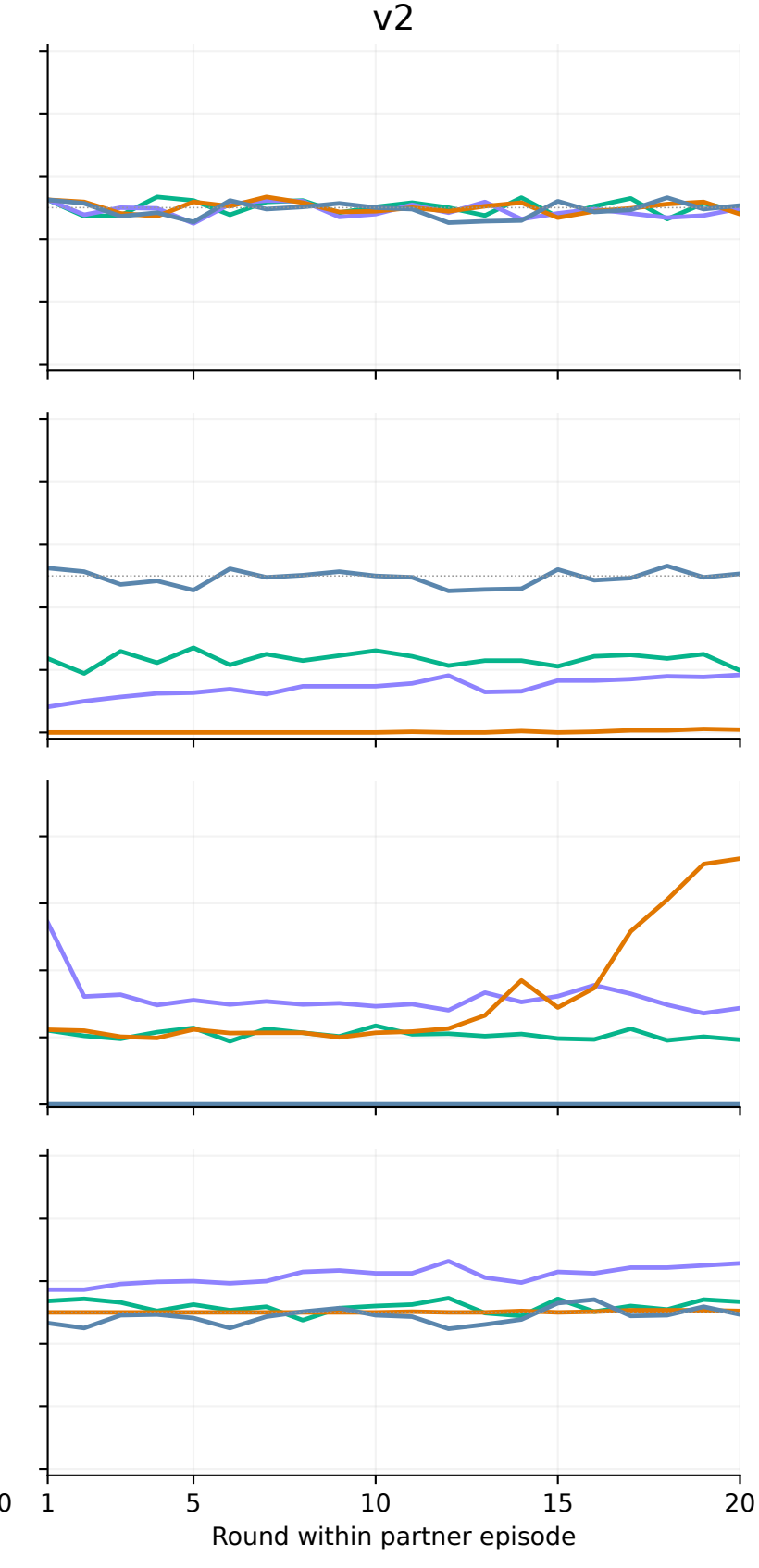
# Assignment behavior: training seed 4

MLP · RNN · RNN single-partner · RNN no influence

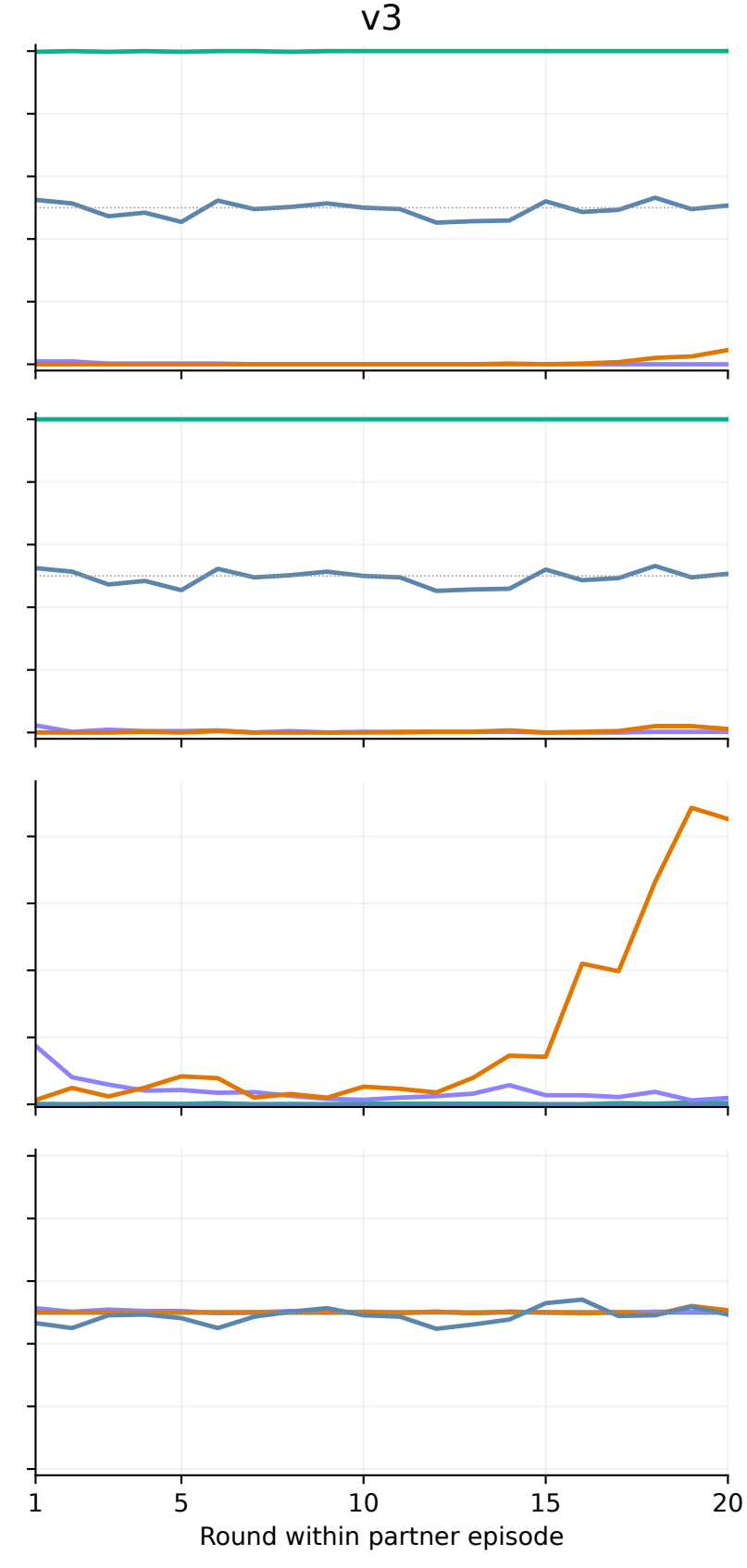
## v1_matched




## v2




## v3

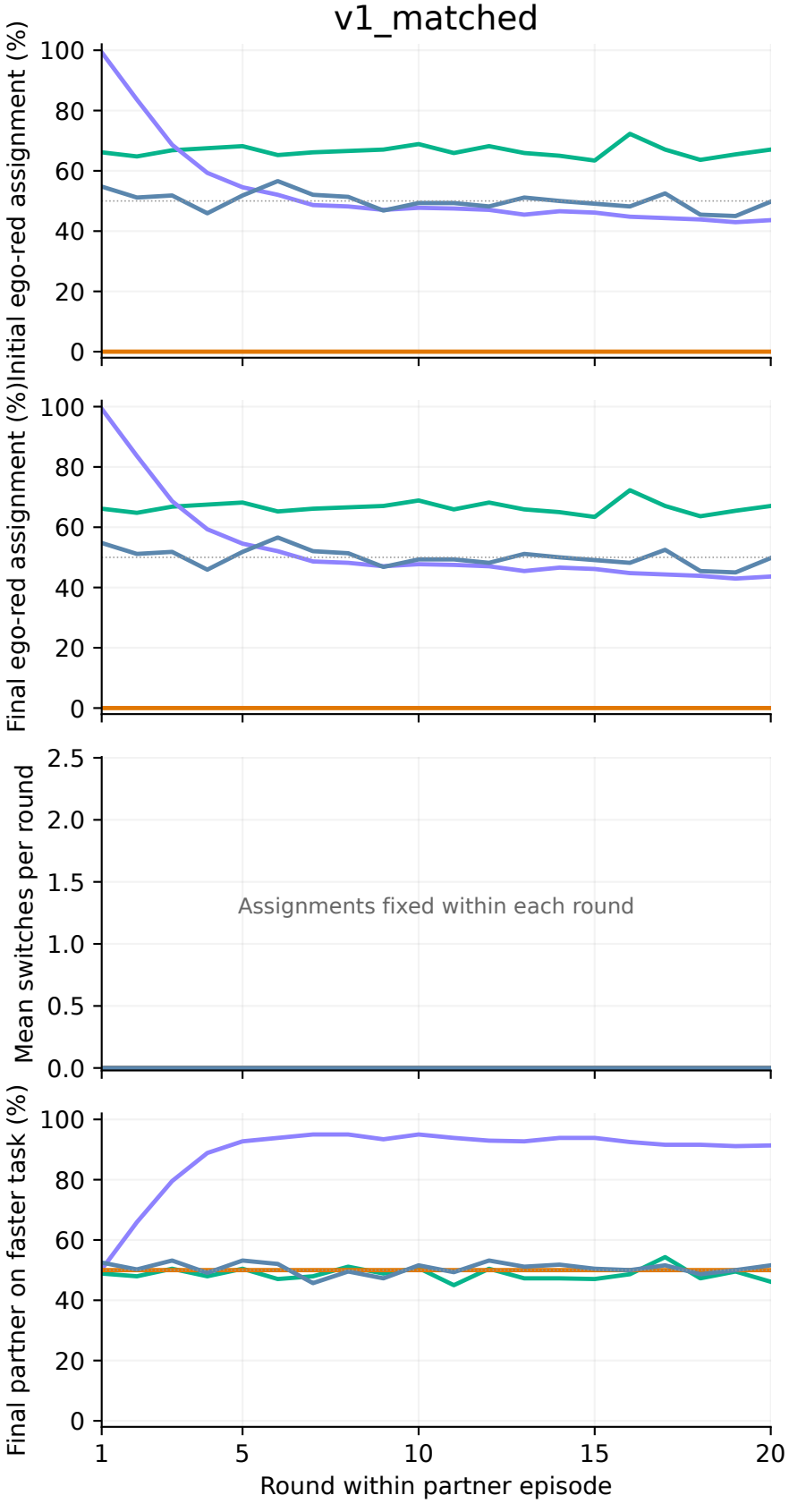



Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

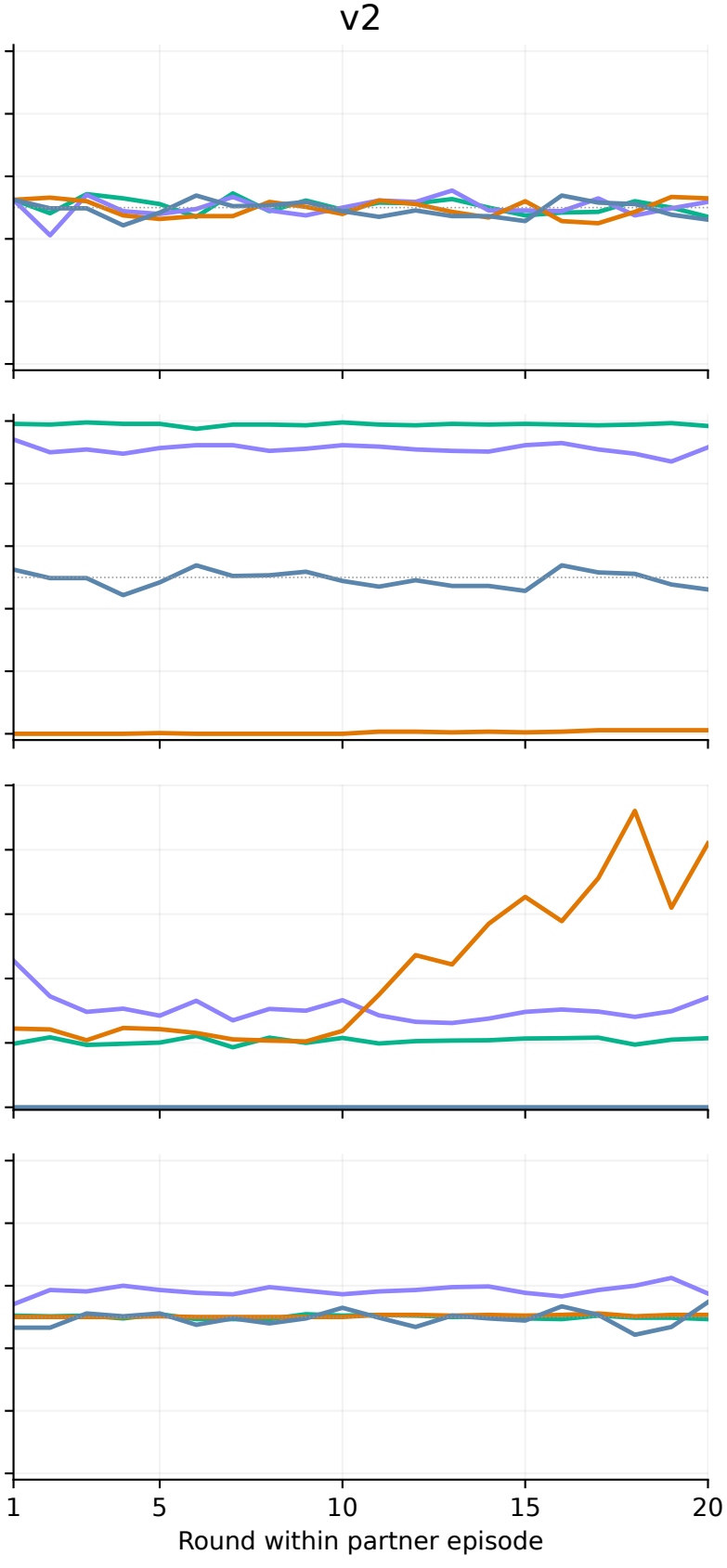
# Assignment behavior: training seed 5

MLP · RNN · RNN single-partner · RNN no influence

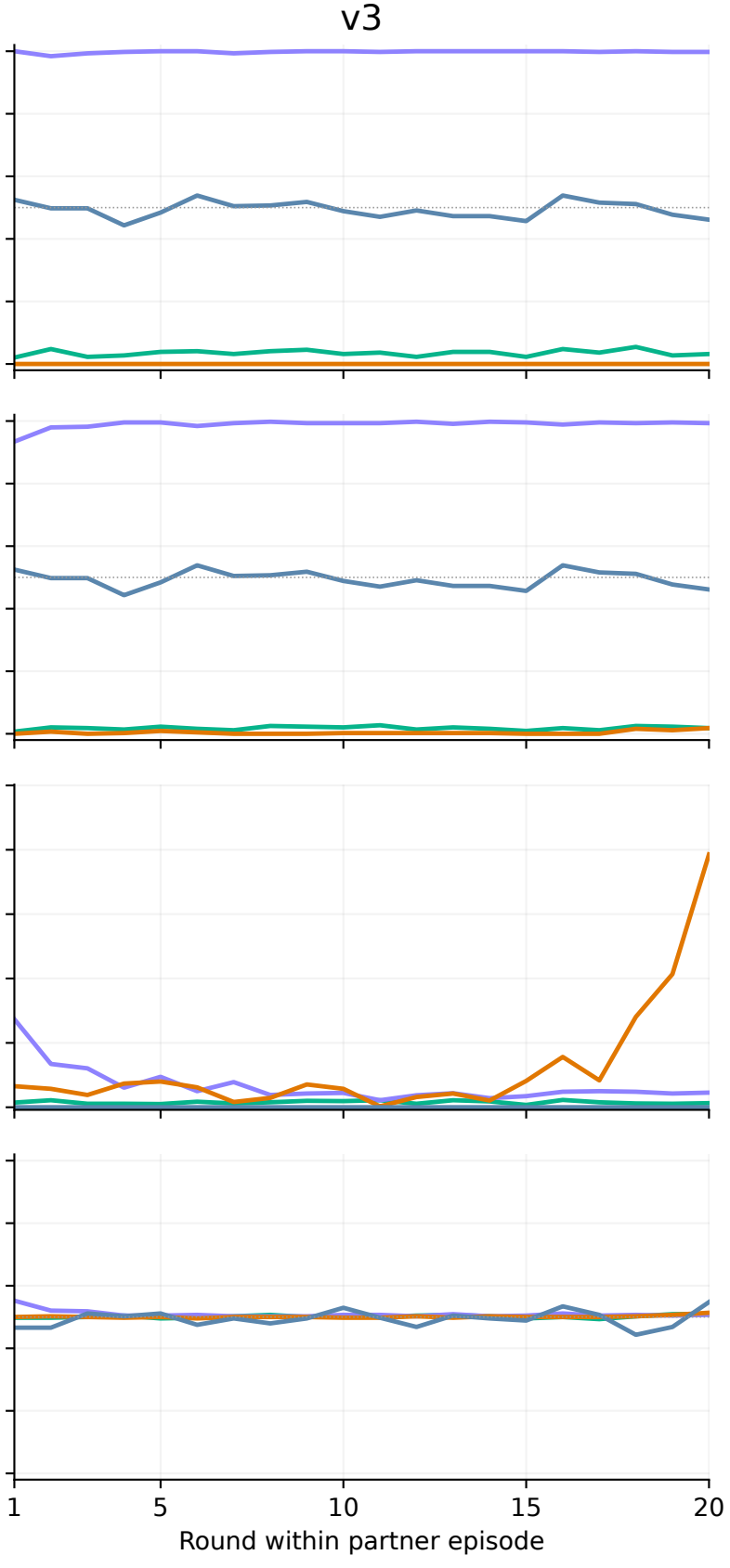
## v1_matched




## v2




## v3




Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.